SQL Online Book Store Analysis

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Basic Queries

1) Retrieve all books in the "Fiction" genre
2) Find books published after the year 1950
3) List all customers from the Canada
4) Show orders placed in November 2023
5) Retrieve the total stock of books available
6) Find the details of the most expensive book
7) Show all customers who ordered more than 1 quantity of a book
8) Retrieve all orders where the total amount exceeds $20
9) List all genres available in the Books table
10) Find the book with the lowest stock
11) Calculate the total revenue generated from all orders

Advance Queries

12) Retrieve the total number of books sold for each genre
13) Find the average price of books in the "Fantasy" genre
14) List customers who have placed at least 2 orders
15) Find the most frequently ordered book
16) Show the top 3 most expensive books of 'Fantasy' Genre
17) Retrieve the total quantity of books sold by each author
18) List the cities where customers who spent over $30 are located
19) Find the customer who spent the most on orders
20) Calculate the stock remaining after fulfilling all order

1) Retrieve all books in the "Fiction" genre.

```
SELECT * FROM Books
WHERE Genre = 'Fiction';
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8
22	Multi-layered optimizing migration	Wesley Escobar	Fiction	1908	39.23	78
28	Expanded analyzing portal	Lisa Coffey	Fiction	1941	37.51	79
29	Quality-focused multi-tasking challenge	Katrina Underwood	Fiction	1905	31.12	100
31	Implemented encompassing conglomeration	Melissa Taylor	Fiction	2010	21.23	44

Books 1

2) Find books published after the year 1950.

```
SELECT * FROM Books
WHERE Published_year > 1950;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
2	Persevering reciprocal knowledge user	Mario Moore	Fantasy	1971	35.80	19
4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8
5	Adaptive 5thgeneration encoding	Juan Miller	Fantasy	1956	10.95	16
6	Advanced encompassing implementation	Bryan Morgan	Biography	1985	6.56	2
8	Persistent local encoding	Troy Cox	Science Fiction	2019	48.99	84

Books 2

3) List all customers from the Canada.

```sql
SELECT * FROM Customers
WHERE country = 'Canada';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wr

Customer_ID	Name	Email	Phone	City	Country
38	Nicholas Harris	christine93@perkins.com	1234567928	Davistown	Canada
415	James Ramirez	robert54@hall.com	1234568305	Maxwelltown	Canada
468	David Hart	stokesrebecca@gmail.com	1234568358	Thompsonfurt	Canada
NULL	NULL	NULL	NULL	NULL	NULL

Customers 3

4) Show orders placed in November 2023.

```
SELECT * FROM Orders
WHERE order_date BETWEEN '2023-11-01' AND '2023-11-30';
```

Result Grid | Filter Rows: | Edit: | Expor

Order_ID	Customer_ID	Book_ID	Order_Date	Quantity	Total_Amount
4	433	343	2023-11-25	7	301.21
19	496	60	2023-11-17	9	316.26
75	291	375	2023-11-30	5	170.75
132	469	333	2023-11-22	7	194.32
137	474	471	2023-11-25	8	363.04
163	207	384	2023-11-23	3	101.76

Orders 4

5) Retrieve the total stock of books available.

```sql
SELECT SUM(stock) AS Total_Stock
FROM Books;
```

Result Grid

Total_Stock
25056

6) Find the details of the most expensive book

```
SELECT * FROM Books
ORDER BY Price DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell C

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
340	Proactive system-worthy orchestration	Robert Scott	Mystery	1907	49.98	88
NULL	NULL	NULL	NULL	NULL	NULL	NULL

7) Show all customers who ordered more than 1 quantity of a book.

```
SELECT * FROM Orders
WHERE quantity>1;
```

Result Grid | Filter Rows: | Edit: | Export/

Order_ID	Customer_ID	Book_ID	Order_Date	Quantity	Total_Amount
1	84	169	2023-05-26	8	188.56
2	137	301	2023-01-23	10	216.60
3	216	261	2024-05-27	6	85.50
4	433	343	2023-11-25	7	301.21
5	14	431	2023-07-26	7	136.36

Orders 7

8) Retrieve all orders where the total amount exceeds $20.

```
SELECT * FROM Orders
WHERE total_amount>20;
```

Result Grid | Filter Rows: | Edit: | Export

Order_ID	Customer_ID	Book_ID	Order_Date	Quantity	Total_Amount
1	84	169	2023-05-26	8	188.56
2	137	301	2023-01-23	10	216.60
3	216	261	2024-05-27	6	85.50
4	433	343	2023-11-25	7	301.21
5	14	431	2023-07-26	7	136.36
6	439	119	2024-10-11	5	249.40

Orders 8

9) List all genres available in the Books table.

```sql
SELECT DISTINCT genre
FROM Books;
```

Result Grid

genre
Biography
Fantasy
Non-Fiction
Fiction
Romance
Science Fiction
Mystery

10) Find the book with the lowest stock.

```
SELECT * FROM Books
ORDER BY stock
LIMIT 1;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Con

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
44	Networked systemic implementation	Ryan Frank	Science Fiction	1965	13.55	0
NULL	NULL	NULL	NULL	NULL	NULL	NULL

11) Calculate the total revenue generated from all orders.

```
SELECT SUM(total_amount) As Revenue
FROM Orders;
```

Result Grid

	Revenue
▶	75628.66

12) Retrieve the total number of books sold for each genre.

```
SELECT b.Genre, SUM(o.Quantity) AS Total_Books_sold
FROM Orders o
JOIN Books b ON o.book_id = b.book_id
GROUP BY b.Genre;
```

Result Grid | Filter Rows:

Genre	Total_Books_sold
Biography	285
Fantasy	446
Science Fiction	447
Mystery	504
Romance	439
Non-Fiction	351
Fiction	225

Result 12

13) Find the average price of books in the "Fantasy" genre.

```
SELECT ROUND(AVG(price), 2) AS Average_Price
FROM Books
WHERE Genre = 'Fantasy';
```

Result Grid

Average_Price
25.98

14) List customers who have placed at least 2 orders.

```
SELECT o.customer_id, c.name, COUNT(o.Order_id) AS
ORDER_COUNT
FROM orders o
JOIN customers c ON o.customer_id = c.customer_id
GROUP BY o.customer_id, c.name
HAVING COUNT(Order_id) >= 2;
```

Result Grid | Filter Rows:

customer_id	name	ORDER_COUNT
84	Gary Blair	2
137	Steven Miller	2
216	Phillip Allen	2
14	John Wood	2
195	Dominique Turner	3
109	Jacob Kelley	2
94	Mr. David Cox	3

Result 14

15) Find the most frequently ordered book.

```
SELECT o.Book_id, b.title, COUNT(o.order_id) AS
ORDER_COUNT
FROM orders o
JOIN books b ON o.book_id = b.book_id
GROUP BY o.book_id, b.title
ORDER BY ORDER_COUNT DESC LIMIT 1;
```

Result Grid Filter Rows: Exp

Book_id	title	ORDER_COUNT
88	Robust tangible hardware	4

16) Show the top 3 most expensive books of 'Fantasy' Genre.

```
SELECT * FROM books
WHERE genre ='Fantasy'
ORDER BY price DESC LIMIT 3;
```

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
240	Stand-alone content-based hub	Lisa Ellis	Fantasy	1957	49.90	41
462	Innovative 3rdgeneration database	Allison Contreras	Fantasy	1988	49.23	62
238	Optimized even-keeled analyzer	Sherri Griffith	Fantasy	1975	48.97	72
NULL	NULL	NULL	NULL	NULL	NULL	NULL

17) Retrieve the total quantity of books sold by each author.

```
SELECT b.author, SUM(o.quantity) AS Total_Books_Sold
FROM orders o
JOIN books b ON o.book_id=b.book_id
GROUP BY b.Author;
```

Result Grid | Filter Rows:

author	Total_Books_Sold
Margaret Moore	8
John Davidson	13
Christopher Fuentes	6
Marissa Smith	16
Christopher Dixon	15
Tonya Saunders	21
Larry Hunt	6

Result 17

18) List the cities where customers who spent over $30 are located.

```sql
SELECT DISTINCT c.city, total_amount
FROM orders o
JOIN customers c ON o.customer_id = c.customer_id
WHERE o.total_amount > 30;
```

Result Grid | Filter Rows:

city	total_amount
Lake Paul	188.56
North Keith	216.60
Kelseyfort	85.50
East David	301.21
Richardsonville	136.36
Ramosstad	249.40
Rogersborough	82.92

Result 18

19) Find the customer who spent the most on orders.

```
SELECT c.customer_id, c.name, SUM(o.total_amount) AS
Total_Spent
FROM orders o
JOIN customers c ON o.customer_id = c.customer_id
GROUP BY c.customer_id, c.name
ORDER BY Total_spent Desc LIMIT 1;
```

Result Grid | Filter Rows:

customer_id	name	Total_Spent
457	Kim Turner	1398.90

20) Calculate the stock remaining after fulfilling all orders.

```
SELECT b.book_id, b.title, b.stock,
COALESCE(SUM(o.quantity),0) AS Order_quantity,
 b.stock- COALESCE(SUM(o.quantity),0) AS
Remaining_Quantity
FROM books b
LEFT JOIN orders o ON b.book_id = o.book_id
GROUP BY b.book_id ORDER BY b.book_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

book_id	title	stock	Order_quantity	Remaining_Quantity
1	Configurable modular throughput	100	3	97
2	Persevering reciprocal knowledge user	19	0	19
3	Streamlined coherent initiative	27	5	22
4	Customizable 24hour product	8	0	8
5	Adaptive 5thgeneration encoding	16	8	8
6	Advanced encompassing implementation	2	0	2
7	Open-architected exuding structure	95	5	90

Result 20

Conclusion

This Online Book Store MIS Analytics Project leverages SQL to transform operational data into actionable business insights. The analysis uncovers strong Fiction and Fantasy performance, a loyal Canadian customer base, and total revenue of $75628.66, primarily from bulk and high-value orders. It also identifies stockouts and underperforming genres as key improvement areas. Overall, the project showcases effective use of SQL for management information reporting—turning raw data into strategic insights that support smarter decision-making and business growth.

Thank You